

Daniel Lu

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Education

Carleton University, B.Sc. in Computer Science, Honours Program (Co-op)

Ottawa, Canada

AI & Machine Learning Stream - Dean's Honours List - CGPA: 3.9/4.0

Sep 2023 – Present

Experience

Blackberry QNX

Jan 2025 – Apr 2025

Software Developer Test Intern

Kanata, Ontario

- Developed tools and automated tests in **Python, C, and Bash** to support **QNX systems** and address **client patch requests**, enhancing **operational efficiency** and improving **testing accuracy**.
- Designed and deployed a **scalable, permanent network testing environment**, enabling **seamless remote access** for development teams across **APAC and EU**, enhancing **global collaboration** and **testing efficiency**.
- Expanded test coverage for **QNX's RTOS** by designing and implementing **automated Pytest-based tests** for the network stack (**io-pkt network manager, packet filter**), ensuring safety certification standards for product release are met.

Software Subteam Lead, Bot Developer, and Mentor

Jun 2021 – April 2024, Part-time

FIRST Robotics Team 8729: Spark Youth Robotics Club

Kanata, Ontario

- **Organized online workshops** tailored toward children to spark interest in **robotics, software development, and engineering**
- Managed the **autonomous and lighting software subteams** during the FRC build season.
- Introduced members to **WPILib, PID systems, and Git workflows** through preparing **weekly lesson plans**.
- Became the **first team from Ottawa** to qualify for the FIRST Championships in **4 years, as rookies**.
- Noticed inefficiencies in managing club tasks across multiple platforms; used **Node.js** paired with **PostgreSQL** to craft a **general-purpose Discord bot** tailored towards streamlining various club tasks.

Projects

CloudQ | [Devpost](#) | [GitHub](#)

Jan 2025, McGill University

- Achieved **Best Web 3.0 Hack** in **McHacks 12** (out of **450+ participants**).
- Deployed an **effective, decentralized, virtual queue management system** that uses **Starknet smart contracts** to enable secure queue creation, joining, and management functionalities.
- Developed a modern, responsive UI using **Vite, React.js, and TailwindCSS**, while integrating **Auth0** for secure, easy user authentication and **Twilio** for SMS notifications.

TRACY: Tennis Realtime Analysis Coaching Systems | [Devpost](#) | [GitHub](#)

Feb 2024, Queen's University

- Achieved **3rd Best Hack** in **QHacks 2024** (out of **200+ participants**).
- Developed an accessible, real-time **React.js web app** for tennis analysis and coaching, implementing **computer vision algorithms** and a **convolutional recurrent neural network** powered by **OpenCV** and **TensorFlow**.
- Effectively tracks **rapid ball and human movements in 3D space** from a **singular viewpoint** and delivers **personalized insights** to enhance the skills of tennis enthusiasts.

Automatic Speech Recognition Neural Network | [Academia](#) | [GitHub](#)

Jun 2022 - Jan 2023, IB Diploma

- Published a machine learning research paper on **automatic speech recognition** that investigated the performances of different **speech feature extraction algorithms** in ASR systems.
- Trained a **deep bidirectional GRU network** using **TensorFlow** to accurately **transcribe real-time continuous speech data** using varying audio feature extraction algorithms (**Mel spectrograms, MFCCs, and discrete wavelet transforms**).

Technical Skills

Languages: C/C++, Python, Java, JavaScript, TypeScript, C#, Rust, SQL, NoSQL, HTML, PHP, CSS

Libraries and Frameworks: TensorFlow, PyTorch, Node, React/Next.js, Angular, Flask, REST APIs, Starknet

Platforms & Technologies: Atlassian tool suite, Git, AWS, Web 3.0, Containerized Systems, Linux/Unix